

Topic:-

Collect metal vessels and various metal articles. Write detailed information. Write the steps in the procedure that can be done in the laboratory for giving glitter to these. Seek guidance from your teacher.

Introduction:-

If you want to polish metals in a laboratory setting, you may need to use specialized equipment and chemicals. Here is a basic procedure for polishing metals in a laboratory:

Clean the metal article: Before you begin polishing, make sure the metal article is clean and free of any dirt, grease, or other contaminants. You can use a mild detergent and warm water to clean the article, then rinse it thoroughly and dry it with a clean towel.

Choose a polishing compound: There are many types of polishing compounds available for use in the laboratory, including diamond paste, alumina, and silica. Choose a compound that is appropriate for the type of metal you are polishing.

Let's understand the method to give shine to individual metals.

✿GOLDEN RING:



◎Prepare a cleaning solution: Mix a small amount of mild dish soap or detergent with warm water in a beaker or a small container.

◎Soak the ring: Place the golden ring in the cleaning solution and let it soak for 10-15 minutes. This will help loosen any dirt or grime on the ring.

◎Scrub the ring: Use a soft-bristled brush or a toothbrush to gently scrub the ring. Pay special attention to any crevices or intricate designs on the ring. Be careful not to scratch the surface of the ring.

⦿Rinse the ring: Once you've finished scrubbing, rinse the ring thoroughly with clean water. Make sure to remove all the soap residue.

⦿Dry the ring: Use a clean, soft cloth to pat the ring dry. Do not use a paper towel or tissue, as they can scratch the surface of the ring.

✿STAINLESS STEEL APPLIANCES:-



⦿Prepare a cleaning solution: Mix warm water with a mild detergent or dish soap in a beaker or a small container.

⦿Dip a soft cloth: Dip a soft cloth or sponge into the cleaning solution and wring it out until it is damp but not dripping.

⦿Wipe the appliance in the direction of the grain: Use the damp cloth or sponge to wipe the stainless steel appliance, following the direction of the grain. Avoid scrubbing too hard or in a circular motion, as this can leave scratches on the surface.

⦿Rinse with clean water: Rinse the appliance thoroughly with clean water to remove any soap residue.

⦿Dry the appliance: Use a clean, dry cloth to wipe the appliance dry, following the direction of the grain. Make sure to remove any water spots or streaks.

✿SILVER LOCKET:-



- ◎Prepare a cleaning solution: Mix warm water with a small amount of mild dish soap or detergent in a beaker or a small container.
- ◎Soak the locket: Place the silver locket in the cleaning solution and let it soak for a few minutes. This will help loosen any dirt or grime on the locket.
- ◎Rinse the locket: After the locket has soaked for a few minutes, remove it from the cleaning solution and rinse it thoroughly with clean water. Make sure to remove all the soap residue.
- ◎Dry the locket: Use a soft cloth to pat the locket dry. Be gentle, as silver is a soft metal that can scratch easily.
- ◎Apply a silver polish: Using a specialized silver polish or cleaner, apply a small amount of the polish onto a soft cloth or sponge. Gently rub the polish onto the surface of the locket, using a circular motion. Be careful not to apply too much pressure, as this can damage the locket.
- ◎Buff the locket: Use a clean, dry cloth to buff the locket, removing any excess polish and giving it a shiny finish.
- ◎Optional: If there are any stubborn tarnish or stains on the locket, you can use a silver polishing cloth or a specialized silver cleaning solution to remove them.

✿BRONZE MEDAL:-



☉Materials:

- ☉Mild dish soap
- ☉Warm water
- ☉Soft-bristled toothbrush or soft cloth
- ☉Lemon juice or vinegar (optional)
- ☉Microfiber cloth or soft towel
- ☉Brass or metal polish (optional)

✿Instructions:

- ☉Fill a bowl with warm water and add a small amount of mild dish soap. Mix well.
- ☉Place your bronze medal in the bowl and let it soak for a few minutes.
- ☉Gently scrub the medal with a soft-bristled toothbrush or soft cloth. Be sure to clean all sides of the medal, including any crevices or engraved areas.
- ☉If the medal is still dirty or tarnished, you can use a mixture of lemon juice or vinegar and water to clean it. Simply soak the medal in the mixture for a few minutes and then rinse it with water.
- ☉If you want to give the medal more shine, you can use brass or metal

polish. Apply a small amount of polish to a soft cloth and rub it onto the medal in a circular motion. Be sure to cover all sides of the medal. Then, use a clean cloth to buff the medal and remove any excess polish.

☉ Once the medal is clean and shiny, rinse it thoroughly with water and then dry it with a microfiber cloth or soft towel.

✿ Advantages:-

1. Improves appearance by giving a shiny, attractive finish.
2. Protects metals from rusting and corrosion.
3. Increases durability and life of the metal objects.
4. Can improve electrical conductivity in some cases.
5. Allows use of cheaper metals coated with expensive ones (cost-effective).
6. Provides smooth and uniform surface finish.

✿ Disadvantages:-

1. The process can be expensive due to chemicals and equipment.
2. Uses harmful chemicals which can cause environmental pollution.
3. Requires proper safety measures in laboratory conditions.
4. If not done properly, coating may peel off.
5. Disposal of chemical waste is difficult and risky.
6. Time-consuming process for thicker coatings.

✿ References:-

1. NCERT Science Textbook (Class 8-10, Metals and Non-metals chapter)
2. Maharashtra State Board Science Book
3. National Council of Educational Research and Training official materials
4. Khan Academy - Electroplating basics
5. BYJU'S - Applications of electroplating
6. School laboratory manual / teacher guidance